

Basic course

Installation

KNX Association

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Preface

This training document is created for the Basic course. Sections with a heading marked [E] contain important information that is also required for the NextGen course. This includes any sub-sections within them. If a heading does not have [E], it means the content is only for reference and is not mandatory for the NextGen course.

The documentation focuses on the most used configuration mode in KNX, i.e., S-Mode (System Mode). Configuration modes that used to be supported in the past like E-Mode (Easy Mode) are not covered further.

1 Safety Extra Low Voltage Networks [E]

Relevant Voltage	Mains Type ¹	Creepage Distance / Clearance	Test Voltage
230 / 400 V	TN / TT	5.5 / 5.5 mm	4.0 kV
400 V	IT	8.0 / 8.0 mm	6.0 kV
24 V	-	1.25 / 0.8 mm	0.6 kV
Ground	-	1.5 / 1.0 mm	1.0 kV

Figure 1: Safety Extra Low Voltage Networks

A Safety Extra Low Voltage (SELV) network is an electrical system designed to operate at low voltage levels, ensuring electrical safety through insulation and separation/isolation from higher voltage circuits.

Key features of a SELV Network:

- Operates at voltages $\leq$ AC 50 V or $\leq$ DC 120 V.
- Electrically isolated from circuits carrying higher voltages (e.g., AC 230/400 V).
- Not connected to earth or protective earth conductors from other circuits.
- Safety transformer is required to maintain safe insulation between SELV and higher voltage systems.

Safety considerations:

- SELV networks minimize the risk of electric shock, as their voltage levels cannot generate lethal currents.
- Voltages below AC 25 V or DC 60 V require no special protection against direct contact in dry environments.

Installation guidelines

- Observe country-specific regulations for both bus and mains installations.
- Maintain separation between SELV wiring and systems carrying higher voltages.
- SELV circuits must also be insulated/isolated from other SELV circuits and ground connections.
- The specified clearance and creepage distances apply under the following conditions:
 - Pollution degree 2 (e.g., office environments).
 - Overvoltage Category 3 (for equipment permanently connected to mains, ensuring high availability).
 - Insulation Material Class 3 (materials with the highest level of insulation).

¹ TN = Terra Neutral, TT = Terra-Terra, IT = Isolated Terra

1.1 SELV – Insulation

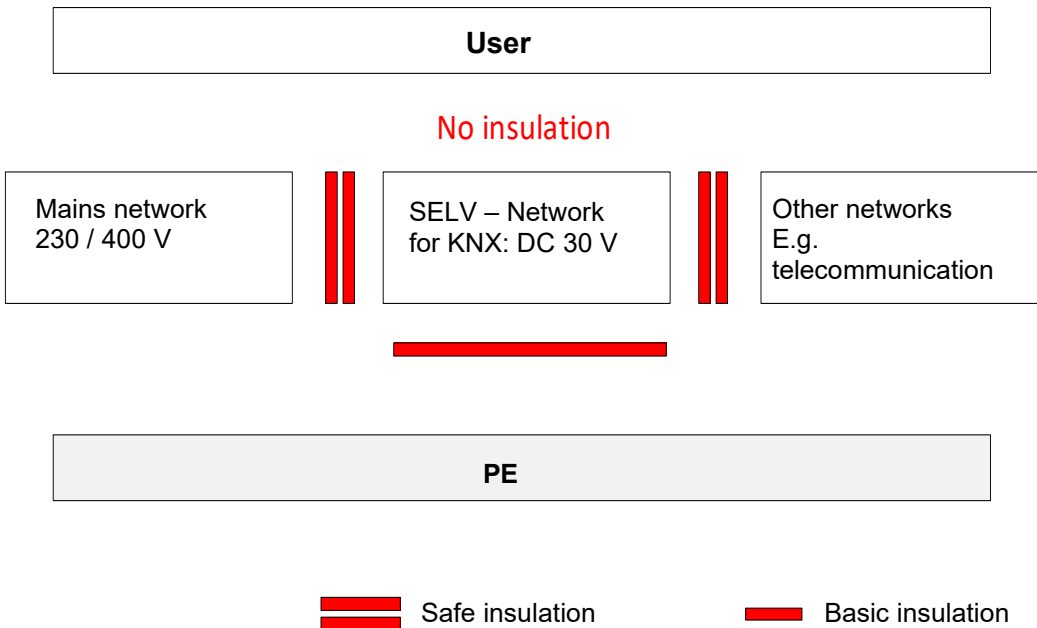


Figure 2: SELV Safety Extra Low Voltage Network

A power supply with secure mains separation is required to generate the SELV voltage for the KNX TP bus.

Voltage used in KNX TP bus

- DC 30 V

Insulation requirements

- Safe insulation from other electrical networks.
- Basic insulation to earth.
- No insulation on the user's side.

Important safety considerations:

- The SELV network shall not be earthed!
- Cables designed for mains voltage networks must not be used for KNX TP networks.

2 Types of bus cables [E]

The YCYM 2x2x0.8 and J-Y (St)Y 2x2x0.8 cables are two common KNX Twisted Pair bus cables, used for fixed installations in building automation. Below is a detailed comparison of their characteristics:

	YCYM 2x2x0.8	J-Y (St)Y 2x2x0.8
Application	• Suitable for fixed installation in dry, humid, and wet environments. • Can be installed wall-mounted, flush-mounted, or in conduits. • Outdoor use is possible if protected from direct sunlight.	• Suitable for fixed installation in dry and humid industrial environments. • It can be wall-mounted, flush-mounted, or installed in conduits. • Outdoor use is only allowed if placed in conduits.
Technical Specifications	• Cable structure: 2 twisted pairs (2x2x0.8 mm) • Test voltage: 4 kV (per EN 50090 standard)	• Cable structure: 2 twisted pairs (2x2x0.8 mm) • Shielding: Foil shielding (St) to reduce electromagnetic interference (EMI) • Test voltage: 2.5 kV (per EN 50090 standard)
Advantages	• Higher test voltage resistance (4 kV), making it more durable in environments with electrical surges. • Suitable for wet environments, making it a better choice for demanding installations.	• Shielded cable reduces electromagnetic interference (EMI), making it ideal for industrial settings. • Suitable for indoor and outdoor applications but must be protected in conduits for outdoor use.

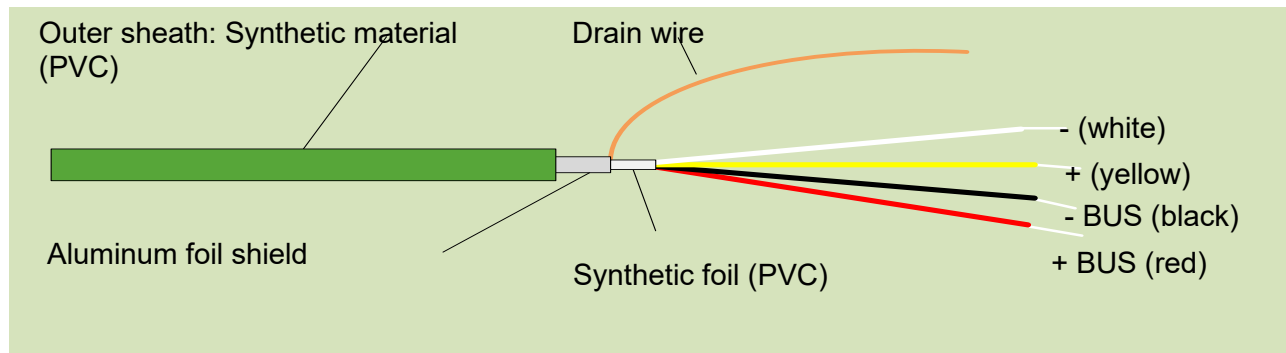


Figure 3: Types of bus cable (J-Y (St)Y 2x2x0.8)

Cables that meet the requirements of Volume 9 of the KNX Specifications (e.g., YCYM 2×2×0.8 or J-Y(St)Y 2×2×0.8) can be certified by KNX Association and may carry the official KNX logo². Only the standardized green KNX TP-certified cable guarantees:

- Reliable data transmission over the maximum line segment length (1,000 m).
- Loop resistance: 75 Ω per 1,000 m.
- Loop capacitance: 100 nF per 1,000 m.
- Maximum distance between two bus devices within a line segment: 700 m.
- Support for up to 256 bus devices per KNX line segment.

For other cable types, the maximum cable length must follow the manufacturer's data sheet to avoid signal loss.

Shielding guidelines:

The shielding of the bus cables shall NOT be connected to the equipotential bonding bar. This ensures that no unwanted currents flow through the bus shielding, which could lead to interference and communication issues.

² For the current list of KNX certified cable types, please consult the KNX web site (www.knx.org)

Wire pair assignments:

- Used wire pair (for KNX TP communication):
 - Red → Plus (+)
 - Black → Minus (-)
- Spare wire pair (white/yellow) - permitted uses:
 - No connection at all (recommended for most installations).
 - For DC SELV power supply (if additional low-voltage power is needed).
 - **NOT** to be used for any modulated signal (e.g., KNX TP communication or voice transmission).
 - Misuse of the spare wire pair for modulated signals can cause interference and communication failures!

Test voltage requirements (EN 50090 Compliance):

- The specified test voltage (4 kV) must be applied to:
 - All connected wire cores (including the drain wire, if present).
 - The outer sheath of the cable.

This ensures the cable's insulation strength and electrical safety, preventing breakdown under normal operating conditions.

Important installation considerations:

- Ensure proper identification and marking of all installed cables to prevent errors during future maintenance or upgrades.
- Follow KNX wiring guidelines to maintain system reliability and interference-free operation.

3 Installation of cables [E]

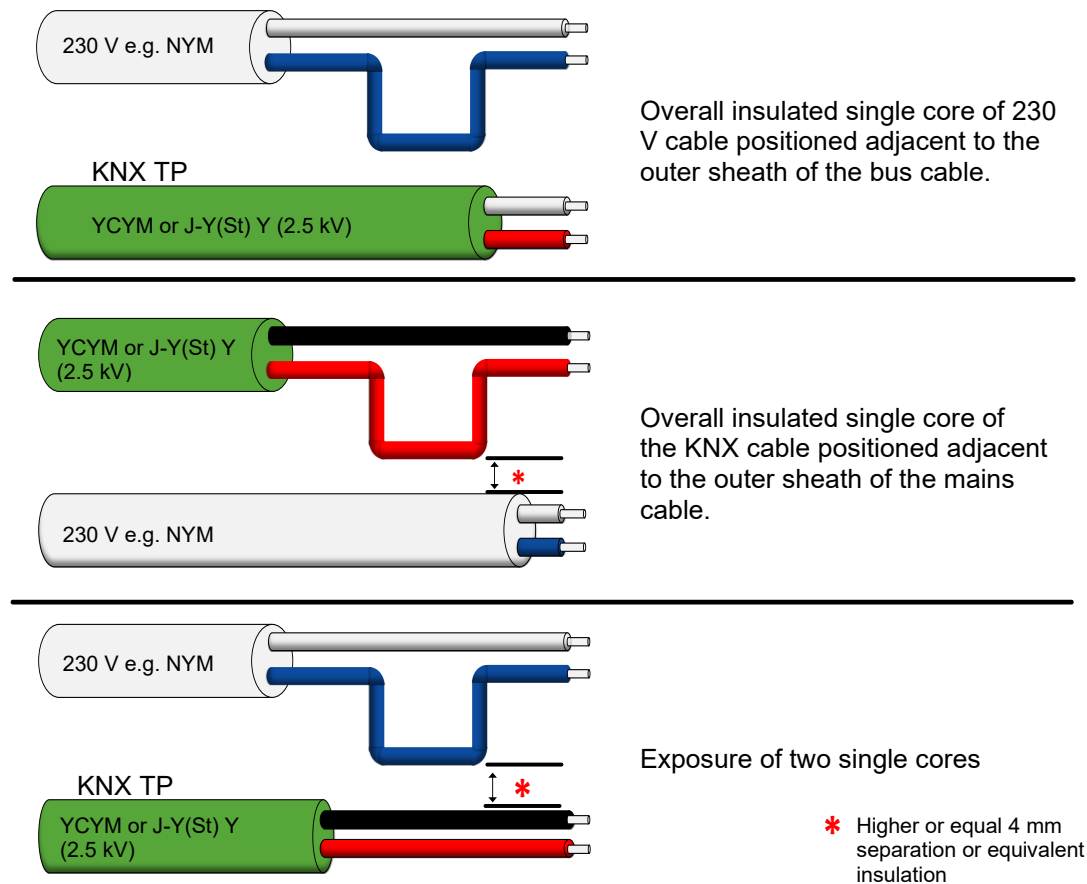


Figure 4: Installation of cables

The installation requirements for KNX bus cables are generally the same as for 230/400 V AC networks, but with some special conditions to ensure compliance and safety.

Special requirements for bus cable installation:

- Parallel installation with mains cables:
 - The KNX-certified YCYM 2x2x0.8 cable may be installed directly next to 230/400 V AC cables due to its higher insulation resistance.
 - The J-Y(St)Y 2x2x0.8 cable may only be installed next to 230 V cables (not 400 V) because of its lower insulation resistance.
- Minimum clearance requirements:
 - A minimum clearance of 4 mm must be maintained between:
 - KNX TP bus cable wire cores and insulated cores of AC 230 V mains cables.
 - Other non-SELV/PELV circuit wires.
 - If 4 mm clearance cannot be ensured, an equivalent isolation must be used, such as:
 - A spacer between cables.
 - Insulation sleeving around wire cores.

- Protection against lightning strikes
 - Ensure adequate distance between KNX TP bus cables and external lightning protection systems (e.g., lightning current arresters).
- Cable marking
 - All KNX bus cables should be permanently labeled as KNX TP or BUS cables for clear identification.

Attention:

- Incorrect polarity in KNX bus cables will prevent the device from functioning until the polarity is corrected.
- A termination resistor³ is **NOT** required for KNX TP networks. The system is designed to function without termination resistors, unlike other bus systems.

³ In signal transmission, electrical signals can reflect when they reach the end of a wire or bus without proper termination. These reflections can cause data corruption, interference, or noise in the system. A termination resistor absorbs the excess energy and prevents these reflections.

4 Bus devices in distribution boards [E]

KNX bus devices can be installed in standardized electric power distribution boards, following specific installation requirements to ensure safe operation and minimal interference with mains power components.

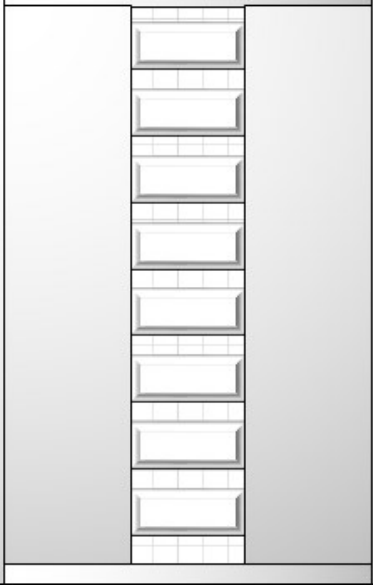
Standardized power distribution boards	Requirements
	- Use of standardized distribution boards. - Install bus cables with sheath up to the bus connector. - Do not install bus devices above mains devices that generate excessive heat.

Figure 5: Bus devices in power distribution boards

Any commercial, EN 50022-compliant distribution board with 35x7.5 mm DIN rails can be used for mounting KNX TP DIN rail devices. If the mains section is separated from the bus installation, no additional special installation requirements apply. If the mains and bus sections are not separated, additional precautions must be taken:

- Sheath bus cables up to the bus connector.
- Prevent contact between mains conductors and bus cable cores through proper wiring and mounting.

Do not mount bus devices above mains devices that generate excessive heat, as this could lead to thermal buildup and affect device performance.

5 Power supply unit

5.1 Introduction [E]

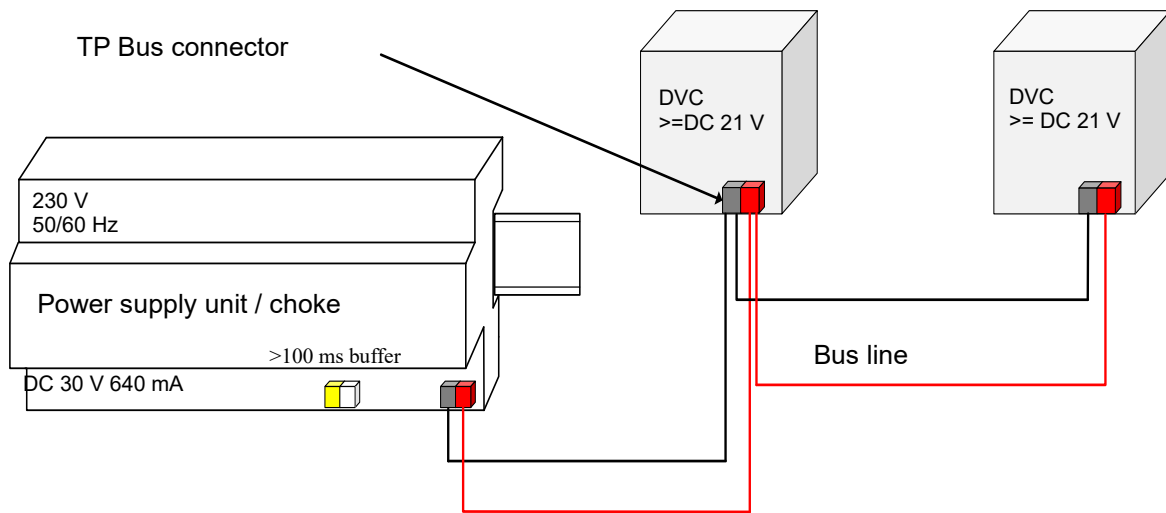


Figure 6: Power supply unit (with TP bus connectors)

Power supply units are essential components in a KNX TP installation, as they generate and regulate the required DC 30 V system voltage.

Key functions of a KNX power supply unit:

- Provides system voltage (DC 30 V) for the KNX TP bus.
- Each KNX line segment requires its own power supply unit to ensure stable operation of all connected bus devices.
- Integrated voltage and current control:
 - Protects against short circuits.
 - Ensures safe and stable power delivery to the bus system.
 - Bridges short power interruptions (minimum 100 ms) to maintain uninterrupted communication.
 - Bus devices require at least DC 21 V for reliable operation.
 - Power consumption of each bus device can be found in the manufacturer's data sheet.

5.2 Power supply in more detail

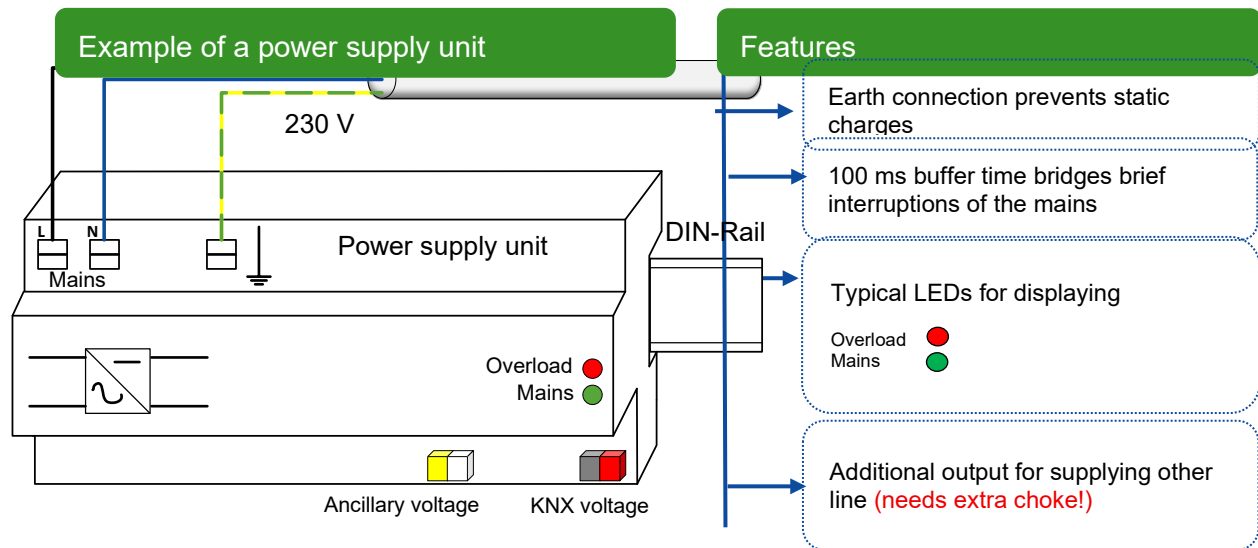


Figure 7: Example and features of a power supply unit

Power supply unit features and functionality

To prevent static charges from accumulating on the bus side, the power supply unit is equipped with high-ohmic resistances that connect each bus core to the PSU's earth connection. To effectively discharge these static charges, this connection must be linked to the protective conductor system, specifically to protective earth of the low-voltage installation. This connection should be marked in yellow/green. However, this does not provide any protective function according to safety regulations and does not conflict with the conditions for SELV networks.

Modern power supplies are equipped with a BCU (Bus Coupling Unit) that provides an electronic reset function, allowing the connected bus line to be reset to 0 V. This reset action can also be remotely controlled via bus telegrams.

Chokes play a crucial role by preventing bus telegrams (AC 9600 Hz signals) from being short-circuited by the DC voltage regulator of the power supply unit.

Types of KNX power supply units

KNX power supplies are available in various output capacities, including:

- 160 mA
- 320 mA
- 640 mA
- 1280 mA

Some power supplies now integrate a BCU, allowing for remote monitoring and reset functions. The number of devices that can be installed on a line segment depends on:

- The power output of the PSU
- The cumulated power consumption of all connected bus devices

Most PSU types include an integrated choke, while some require an additional external choke.

Installation and additional features

Most modern power supply units are DIN rail mounted, with bus voltage accessible via the bus connector. Some models include an ancillary voltage output, which allows additional lines to be powered using a separate choke.

Additional features available in some PSU models include:

- Uninterruptible power supply (UPS) functionality
- Floating relay outputs for detecting and signaling mains failure or normal operation
- LED indicators to display the operating status

LED indicator meanings:

- **Green:** The power supply is active and functions within its normal operating range.
- **Red:** The power supply is overloaded, possibly due to a short circuit in the bus wiring or excessive current draw from connected devices.

6 Power supply for two line segments

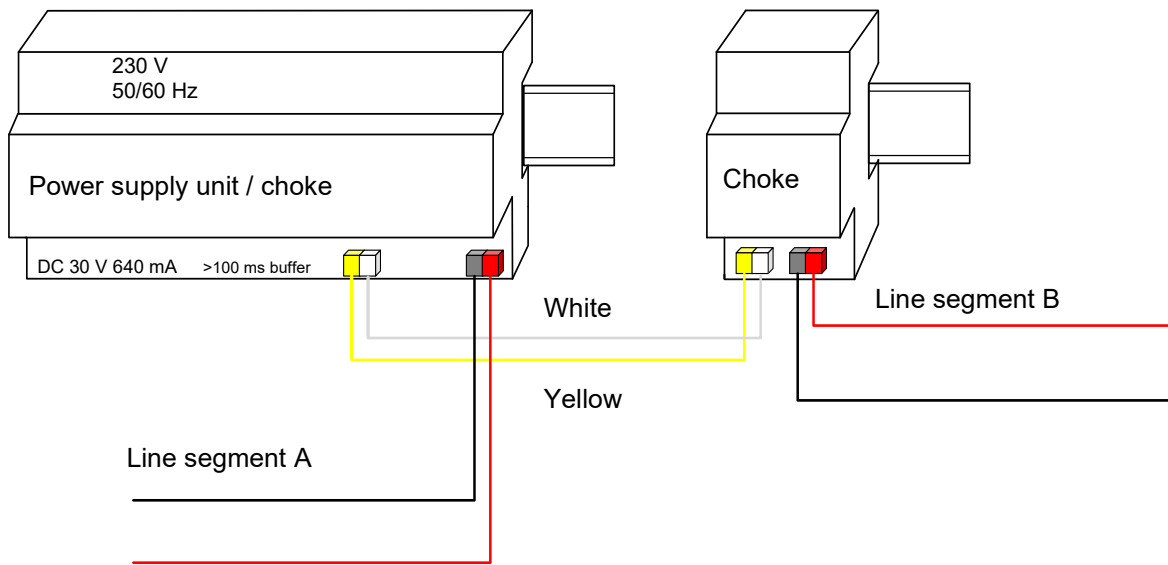


Figure 8: Power supply for two line segments

In KNX installations, when additional current is required, a single power supply unit may be used to feed two separate KNX lines. This setup depends on the total load and the capabilities of the power supply unit. Each KNX line must have its own choke to ensure proper bus communication. Some power supply units have an integrated choke only for one line, requiring an additional external choke for the second line.

Bus voltage and ancillary power connections

- Bus Voltage
 - Dark grey/red bus connector: Provides (choked) DC 30 V KNX bus voltage to power KNX bus devices.
- Ancillary power
 - White/yellow connector: Supplies additional power to the second line via a separate choke.

6.1 Two power supply units in one line segment

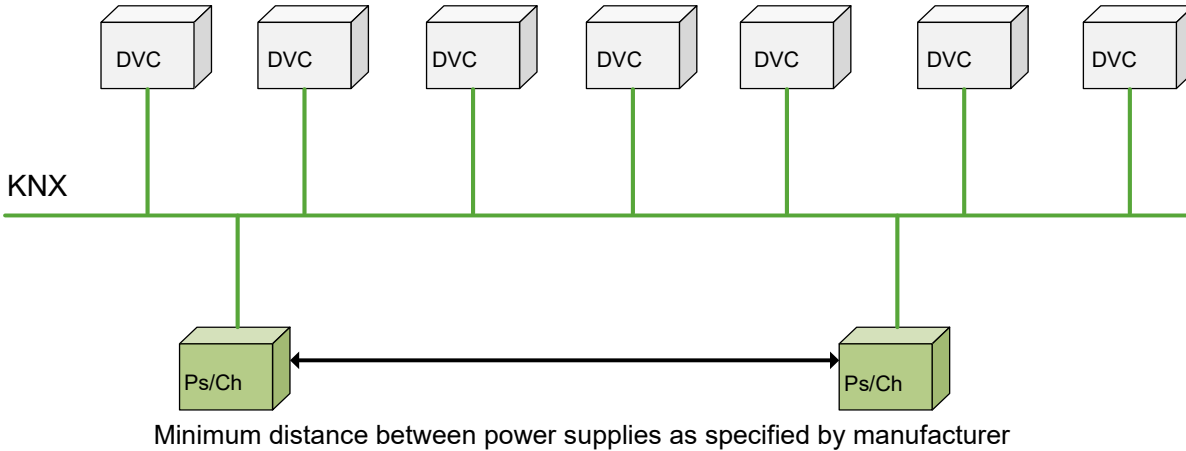


Figure 9: Two power supply units in one line segment

If e.g., more than 30 bus devices are installed in close proximity (e.g., in a distribution board), the PSU should be installed nearby to reduce voltage drop and improve power distribution. If an additional power supply unit is required in the same line segment, the **minimum distance between the two PSUs must follow the manufacturer's specifications.**

It is not allowed to have more than two power supply units in one line segment.

8 Bus cables in wall boxes [E]

When installing KNX bus cables in wall boxes, strict SELV insulation requirements must be followed to ensure protective separation from mains voltage circuits.

SELV circuits (KNX TP bus cables) must have double or reinforced insulation between mains (AC 230 V/400 V) and bus cables. Unsheathed wire cores must never come into contact with mains cables.

The cables can be installed in two ways:

	Separate wall box: • A dedicated wall box is used only for KNX bus cables, separate from mains wiring.
	Shared wall box with partition: • A shared wall box can be used only if a partition is installed. • The partition must ensure a minimum clearance of 5.5 mm between bus cables and mains circuits. • The same distance applies for creepage clearance in TN/TT networks (common in office buildings).

9 Installation of flush-mounted bus devices [E]

When installing flush-mounted KNX bus devices, specific wall box and insulation requirements must be followed to ensure safe and compliant installation.



- Use of wall boxes for screw mounting
- Permitted use of flush-mounted devices in combination with mains devices depends on the environmental conditions and the design of the bus devices (e.g. pollution degree, overvoltage category).

Figure 10: Installation of flush-mounted bus devices

Wall box requirements:

- Screw mounting only
 - Only wall boxes suitable for screw mounting may be used.
 - Clamp-mounted wall boxes are generally not suitable for KNX devices.
- Recommended wall box depth
 - To ensure adequate space for bus cables, use wall boxes with a minimum depth of 60 mm.
 - This prevents cable congestion and makes installation easier.

Installing bus devices with mains devices ("Combinations"):

- "Combinations" refer to installing both mains-powered devices (e.g., power sockets) and bus devices (e.g., KNX push buttons) under a common cover.
- Safety and isolation requirements:
 - Mains devices and bus devices must be safely isolated from each other.
 - Basic insulation is required for mains devices.
 - Basic insulation of bus devices is required against 230 V.

Attention:

- Before installing a KNX bus device with mains devices, check the manufacturer's data sheet whether such an installation is allowed.
- Some KNX devices have specific installation requirements, such as connecting the frame to the protective earth conductor.
- Even if the common box cover is removed, mains devices must not be accessible to prevent accidental contact.

10 Standardized TP Bus connector [E]

The KNX TP bus connector is a standardized component used for connecting, branching, and extending KNX bus cables while ensuring reliable operation and safety.



Figure 11: Bus connector

Structure of the bus connector:

- The KNX bus connector consists of two mechanically linked parts (via a dovetail joint):
 - Plus terminal (red)
 - Minus terminal (dark grey)
- The bus connector supports up to 4 bus wires (6 mm stripped length) per part, using screwless terminals for easy installation.
- The bus connector allows removal of bus devices without interrupting the entire bus line, ensuring continuous operation.
- The bus connector ensures mechanical protection against reverse polarity, reducing installation errors.

Functions of the KNX TP bus connector:

- **Branching the bus cable** to connect multiple devices.
- **Extending the bus cable** to accommodate longer distances.
- **Protecting the ends of the bus cable** to prevent damage or accidental contact.
- **Connecting bus devices** to the KNX TP network.

A bus cable should only terminate at either the KNX device itself or at a KNX bus connector. To avoid confusion with other electrical circuits, the bus connector should only be used for KNX TP networks.

The standardized TP bus connector is the most used bus connector in KNX systems. However, KNX allows manufacturers to use other types of connectors as long as they meet the minimum requirements for connectors specified in Volume 9/1 of the KNX Specifications.

11 Lightning protection measures (Primary Protection)

The KNX TP bus network must be integrated into the lightning protection system of the mains power network to ensure safety and compliance with IEC 62305-3.

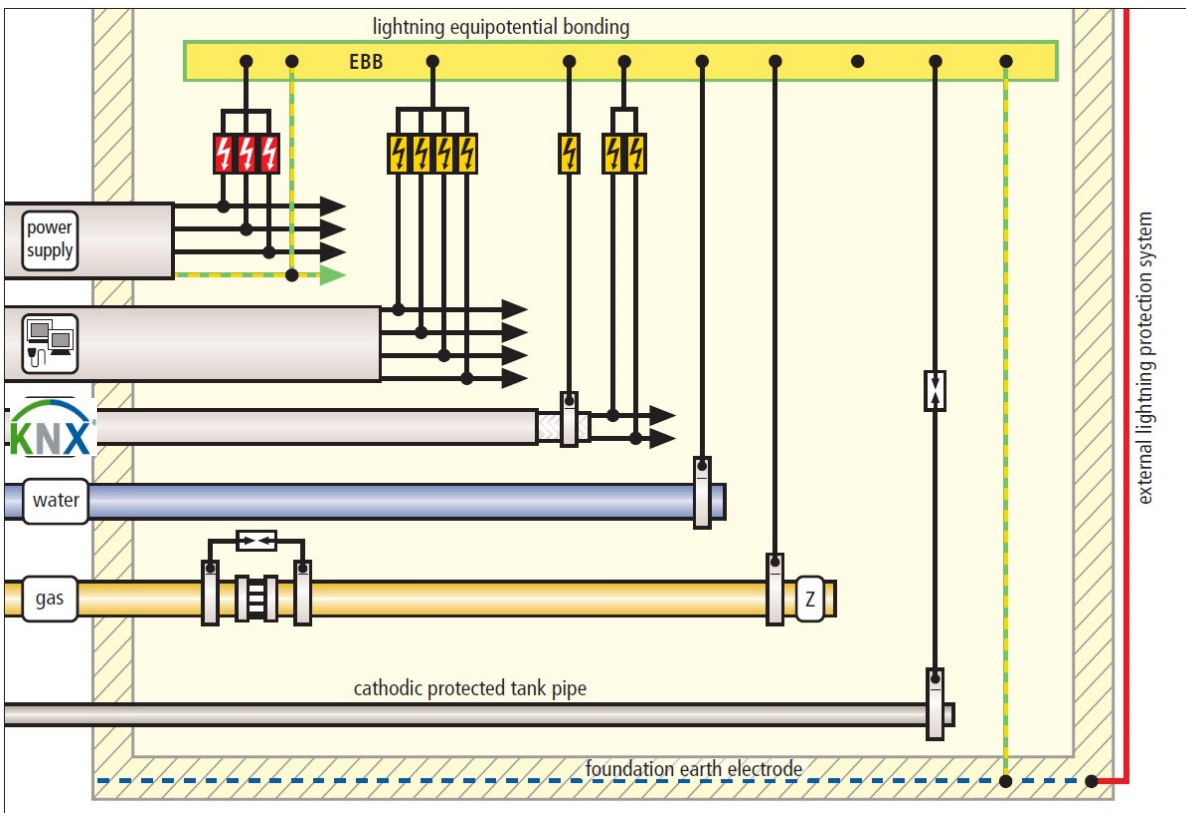


Figure 12: Lightning equipotential bonding according to IEC 62305-3 (www.dehn.de)

When is lightning protection required?

Lightning protection measures may be necessary due to:

- Local building regulations mandate protection.
- Risk analysis of the building based on IEC 62305.
- Insurance company requirements for high-risk buildings.

Buildings particularly vulnerable to lightning strikes or severe damage include:

- Public buildings (e.g., hospitals, conference centers, government offices).
- Industrial and commercial facilities.
- Tall structures or isolated buildings.

Internal lightning protection (Equipotential Bonding):

Internal lightning protection is a critical component of any lightning protection system.

- Lightning protection equipotential bonding bar
 - All conducting systems (water pipes, gas pipes, metal structures, heating systems) must be connected to an equipotential bonding bar.
 - Active conductors must be indirectly connected via lightning current arresters (SPD devices).
 - This method is known as "Primary Protection", as it ensures controlled energy dissipation in case of a lightning strike.

Primary protection measures:

- For AC 230/400 V mains protection
 - Nominal discharge current: $\geq 12.5 \text{ kA}$ (10/350 μs) per conductor
 - Protection level: $< 4 \text{ kV}$
 - Surge Protection Device (SPD) Type 1 according to IEC 61643-11:2012
- For KNX TP bus line protection
 - Nominal discharge current: $\geq 2.5 \text{ kA}$ (10/350 μs) per conductor
 - Protection level: $< 600 \text{ V}$
 - Surge Protection Device (SPD) Type 3 according to IEC 61643-11:2012

12 Protection measures for bus cables installed between buildings

When bus cables extend across multiple buildings, special lightning and surge protection measures must be taken, even if a lightning protection system is not installed. These measures protect KNX installations from damage due to lightning strikes and transient surges.

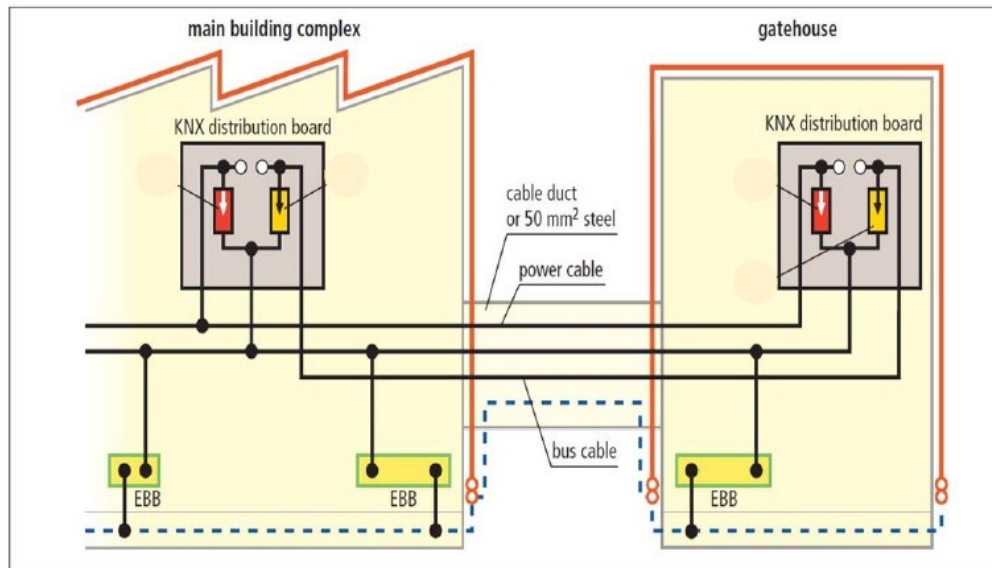


Figure 13: Lightning and surge protection for cabling systems installed between buildings with interconnected foundation earth electrodes (www.dehn.de)

Two options:

1. Installation of a lightning current arrester at the building entrance:
 - Install a lightning current arrester at the next corner of the building.
 - The arrester must be connected to the main equipotential bonding.
2. Use of steel cable ducts.
 - Bus cables should be routed in a steel cable duct when installed between buildings.
 - The steel duct must be grounded at both ends (i.e., at the building entrances).
 - To discharge parts of the lightning current, the grounding conductor must have a minimum cross-section of:
 - * 16 mm² Copper (Cu)
 - * 25 mm² Aluminum (Al)
 - * 50 mm² Iron (Fe)
 - These values follow the IEC 62305-3 standard.
 - This acts as a shielding measure, reducing the risk of induced surges.

Secondary overvoltage protection for bus devices:

- Use overvoltage protectors in each building
 - Bus devices together with the line couplers inside the building should be connected to an overvoltage protector.
- Maintain distance between bus devices and overvoltage protectors
 - Bus devices and the overvoltage protector should be installed several meters apart.
 - This prevents the overvoltage protector from being subjected to primary surge currents, allowing it to perform its function effectively.

13 The prevention of loops

In KNX TP installations, loop formation must be avoided as it can lead to major surges during lightning strikes, causing flashovers in bus devices. The larger the loop area, the higher the induced peak surge, which increases the risk of system failures.

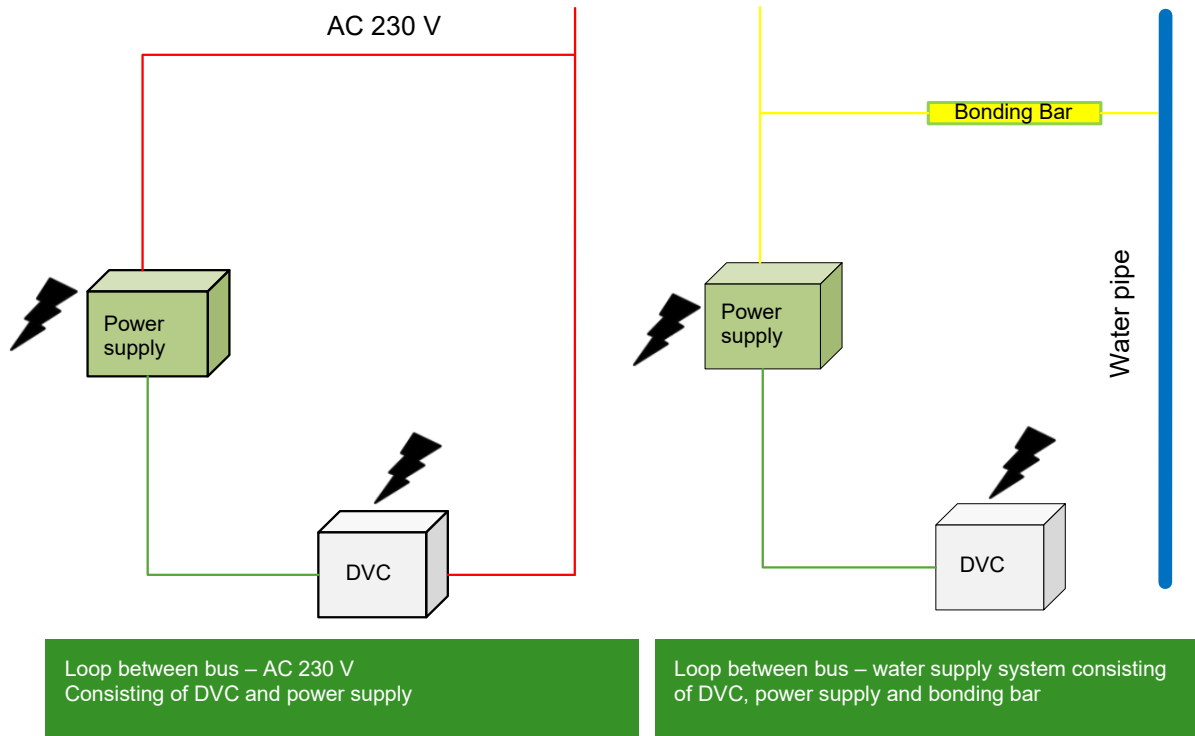


Figure 14: Prevention of loops

How loops are created in KNX installations:

- Loops occur when e.g., a KNX power supply unit is connected to both the bus network (SELV) and:
 - the 230 V mains network.
 - To metallic structures such as the water supply system, central heating system, metal walls or frames via the equipotential bonding bar.

How to prevent loops in KNX installations:

- Plan the installation layout properly
 - Prevent loop formation during the design phase by carefully planning cable routing.
- Install bus and mains cables close together
 - Minimize separation between KNX bus cables and 230 V mains cables to reduce loop area.
- Maintain distance from metal structures
 - Keep bus and mains cables away from metallic infrastructures (e.g., pipes, walls, bonding bars).

If line-crossing loops occur in a KNX TP installation, it may under certain circumstances not be possible to properly program the installation.

14 Basic immunity of bus devices

The basic immunity of KNX bus devices is tested according to IEC 63044-3, ensuring protection against common surge voltages caused by switching operations in electrical systems.

Standardized surge protection for bus devices:

- **Surge Test:** 2 kV surge voltage (core to earth) is applied during testing.
- **Purpose:** Ensures bus devices are resistant to surges caused by switching operations in electrical networks.
- **Result:** This level of protection is sufficient in most standard installations.

Potential causes of stronger Electromagnetic Interference (EMI)

Despite the built-in immunity, significant interference may still occur in the following cases:

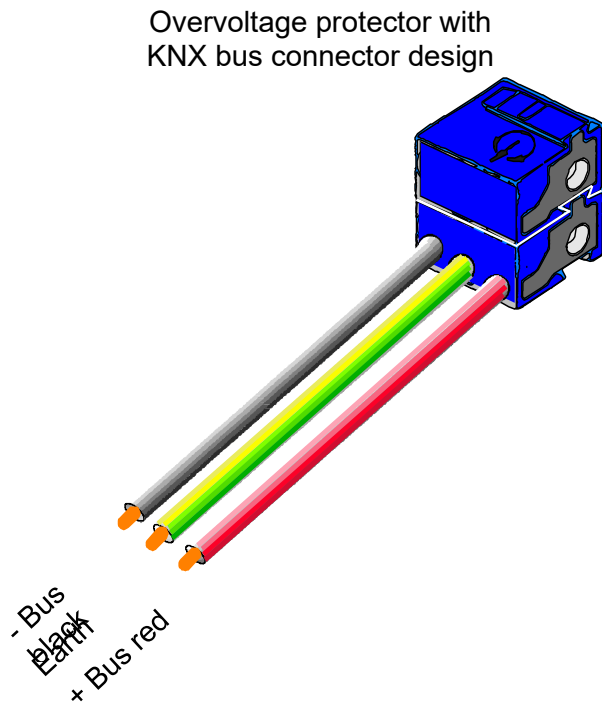
- **Parallel installation of bus and high-power mains cables**
 - When KNX bus cables are installed alongside high-power mains cables over long distances, strong electromagnetic fields can induce voltage surges in the bus system.
- **Proximity to lightning rods and arresters**
 - If KNX bus cables are installed near lightning rods or surge arresters, they may be exposed to high surge currents from lightning strikes.
- **Bus cables running parallel to conductive structures**
 - If bus lines are installed next to conductive parts of a system (such as metal structures, grounding cables, or pipes that carry lightning current), they may pick up induced voltage surges.
- **Formation of loops**
 - Bus loops (unintended closed circuits) can act as antennas, increasing susceptibility to electromagnetic interference (EMI).
 - Loops should be avoided during installation (see Section 13).
- **Bus devices connected to conductive materials**
 - If KNX devices are physically connected to conductive parts (such as metal walls, heating pipes, or grounding systems), they may be directly affected by potential surges.

15 Bus devices on cable ends

Bus devices installed at the end of a KNX TP bus cable are more susceptible to overvoltage surges and electromagnetic disturbances. To ensure reliable operation and long-term protection, additional secondary protection measures should be implemented.

16 The overvoltage protector with KNX bus connector design (Secondary protection)

The overvoltage protector is a secondary protection device designed to protect KNX bus devices from electrical surges, particularly those caused by lightning or switching operations.



Recommended usage

- To bus devices with 230 V mains connection
- To backbone/line/segment couplers or line repeaters on all TP sides
- To bus devices installed in conductive walls or near metal structures (e.g., water pipes, gas pipes, heating pipes etc.)
- At the edge of buildings
- In case of large surface loops with mains

Figure 15: The overvoltage protector with KNX bus connector design

Recommended usage of overvoltage protectors:

Installation inside distribution boards:

- One overvoltage protector per TP line segment i.e., to backbone/line/segment couplers or line repeaters on all TP sides. (Or: Add overvoltage protectors near all power supplies)
- Make sure that the outer conductor (L) and neutral conductor (N) are also equipped with appropriate overvoltage protectors for 230/400 V.

Installation outside of distribution boards:

- Bus devices with Protection Class 1 (which require grounding).
- Bus devices with additional connections (e.g., 230/400 V mains or heating system pipes).
- In all places where bus and mains wiring create large-surface loops, increasing the risk of surge exposure, e.g.,
 - at decentralized (shutter) actuators
 - near luminaires with built-in switching actuators
- In all places where bus and earth potential wiring create large-surface loops, e.g.,
 - in conductive walls,
 - near earthed water pipes, gas pipes, heating pipes

Overvoltage protector requirements:

- Nominal discharge current: $\geq 5 \text{ kA}$ (8/20 μs)
- Protection level: $< 350 \text{ V}$
- KNX certified

Design and functionality of the overvoltage protector:

- **Symmetrical safety design**
 - The overvoltage protector discharges both bus wires (red and black), preventing large voltage differences.
 - Single-pole arresters are not allowed, as they do not provide balanced protection.
 - Varistors are also not suitable due to their higher capacitance, which may interfere with bus communication.
- **Bus connector integration**
 - The overvoltage protector features connection wires that match KNX bus cable colors (red = +, black = -).
- **It can be connected to flush-mounted bus devices and couplers:**
 - Directly to a bus device (for flush-mounted installations).
 - To the KNX bus by using a conventional KNX bus connector.

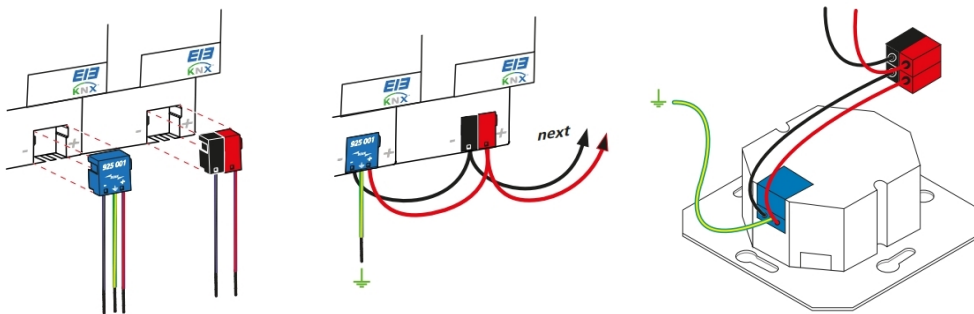


Figure 16: Connection overvoltage protector

- **Protective Earth connection**
 - A third wire (green/yellow) serves as the earthing conductor.
 - This must be connected to the nearest protective earth (PE) point in the installation.
- **Distribution board connection**
 - The earthing conductor in the distribution board must be connected to the PE conductor using a **non-earthed** DIN rail terminal.

Note: The overvoltage protector cannot be used to branch the bus cable.

17 Checking the installation

To ensure a fully functional and compliant KNX TP installation, a systematic inspection must be performed before commissioning.

Key inspection steps:

1. Verify cable length limits.

Confirm that all bus cable lengths comply with the maximum allowed distances:

	Maximum Allowed Distance
Length of a line segment	1,000 m
Distance between power supply unit and bus device	350 m
Distance between two power supply units (including chokes)	As specified by manufacturer
Distance between two bus devices	700 m

Voltage drops and increase of the transmission duration of telegrams are the result of ohmic resistance, capacity, and inductance of bus cables.

2. Verify bus cable identification.

- Ensure that bus cable ends are clearly marked as KNX TP or BUS.
- Indicate the area and line address for easier maintenance and troubleshooting.

3. Check for incorrect cable connections.

- Bus cables from different lines must never be connected together.
- Tests for inadmissible connections can be carried out by:
 - Switching off the power supply of the line under test.
 - Checking if the line coupler's power LED remains lit - this indicates an unwanted connection.

4. Measure insulation resistance of bus cables.

- Perform an insulation resistance test at DC 250 V.
- The insulation resistance must be at least 500 kΩ.
- Test method: Measure conductor against PE, not conductor against conductor.
- **Note:** Remove overvoltage protectors before testing to prevent damage and inaccurate readings.

5. Perform polarity check for all bus devices.

- Set each bus device into programming mode by pressing the programming button.
- If the programming LED lights up, the polarity is correct.
- Press the button again to exit programming mode and reset the LED.

6. Measure bus voltage at cable ends.

- After all bus devices are installed, measure the bus voltage using a voltmeter.
- The measured voltage must be at least DC 21 V.

7. Record all test results.

- Document the test results and add them to the KNX TP installation records.

8. Hand over ETS project and documentation to client

- Provide the ETS project file and installation documentation to the client.
- Obtain a signed agreement from the client, confirming:
 - All implemented functions meet expectations.
 - The installer's warranty expires if modifications are made to the realized KNX installation.